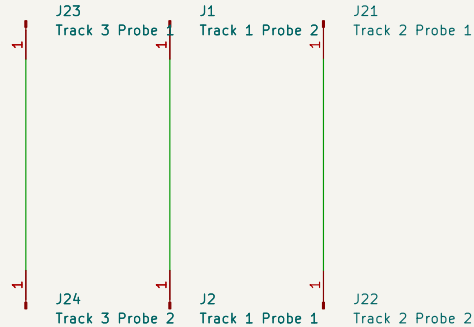


### PCB Track Resistance



#### 4-Wire Kelvin ACIR Measurement:

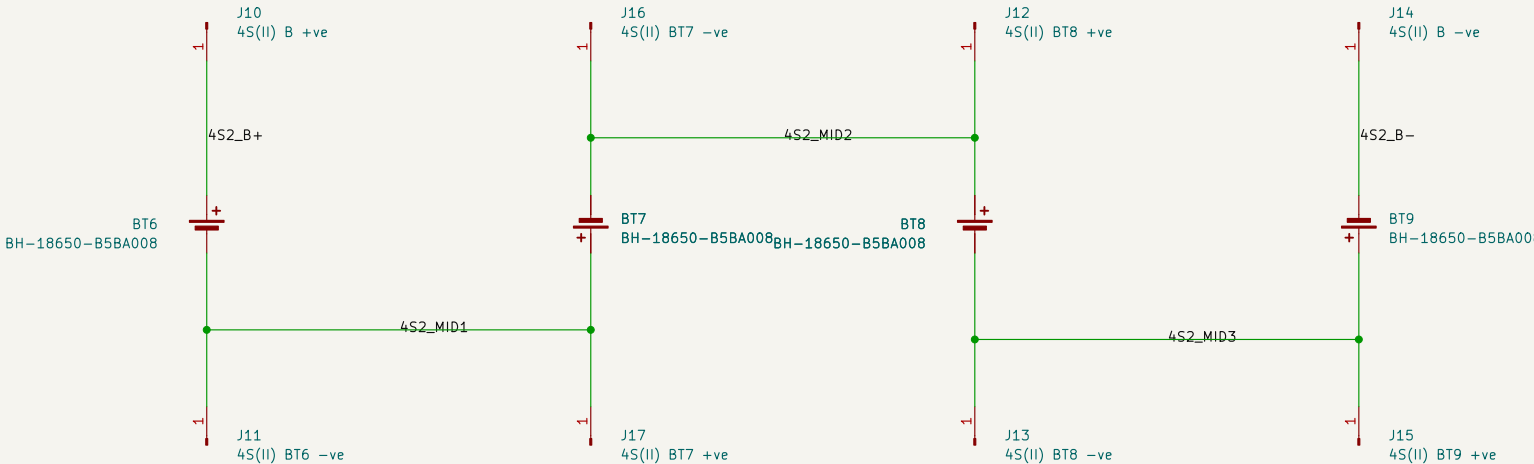
Inject 1 kHz AC current across B+ and B- terminals.

Sense AC voltage drop to isolate purely ohmic internal cell resistance.

### 4S 1P MODULE (Series String – Voltage Stacking & 4x ACIR)

#### 4S 1P (II) MODULE (Varying Interconnect Track Resistance – Educational Pack Experiment)

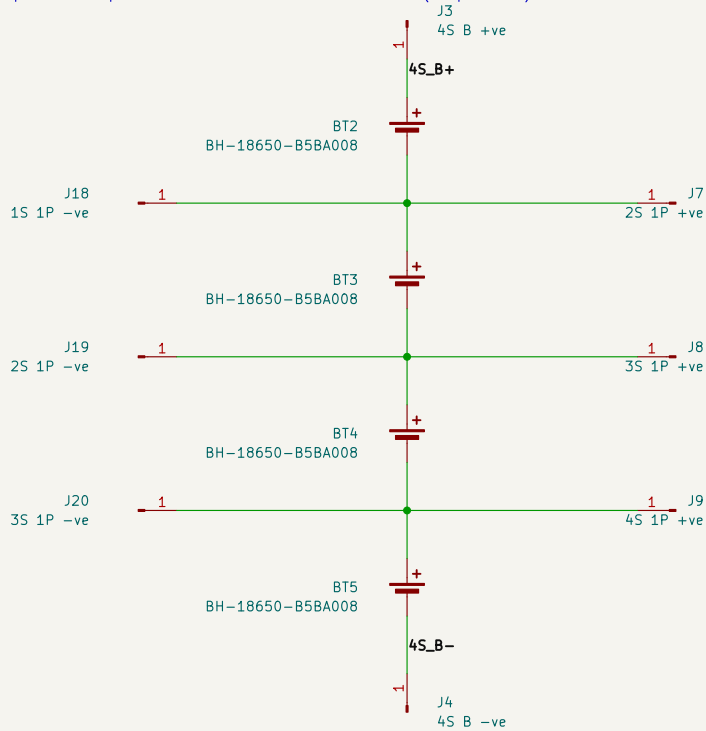
Current Distribution & Branch Kelvin Sensing:  $1/R_{eq} = 1/R1 + 1/R2 + 1/R3 + 1/R4$  ( $R_{eq} \approx R/4$ )



Cells have IDENTICAL ACIR (~20mOhm); Series interconnects vary systematically:  
Link 1 (BT6–BT7): Ultra–wide 10mm 2–layer bus with dense via matrix (~0.5mOhm)  
Link 2 (BT7–BT8): Medium 3.0mm single–layer trace (~3.0mOhm)  
Link 3 (BT8–BT9): Narrow 0.8mm single–layer trace (~10.0mOhm)  
Demonstrates how interconnect trace impedance directly inflates total pack ACIR.

### 4S 1P MODULE (Series String – Voltage Stacking & 4x ACIR)

Total Series Impedance:  $R_{eq} = R1 + R2 + R3 + R4 + R_{interconnect}$  ( $R_{eq} \approx 4 * R$ )



Inter–cell Kelvin probing pairs (J18/J7, J19/J8, J20/J9) allow measuring:

- Individual cell ACIR in–situ under series string current loading.
- Interconnect contact resistance across cell–to–cell busbars.

### MECHANICAL MOUNTING HOLES

8x M3 Plated Chassis Grounding / Standoff Fixtures



### EIS DESIGN & FABRICATION SPECS

#### 1) Kelvin 4-Wire EIS Architecture:

Separate excitation and sensing paths eliminate test lead resistance.

#### 2) Ultra-Dense Micro-Via Array Stitching:

1,792 stitched vias across high-current copper planes.

Effective parallel capacitance  $C_{eff} < 0.001$  pF (negligible).

Halved DC resistance ( $R_{dc} / 2$ ) for maximum measurement SNR.

#### 3) PCB Stackup & Layer Specs:

2-Layer 1.6mm FR4, 1oz Cu, Matte Black Mask, HASL Lead-Free.

|                                                                                                                                                                                                                                                                                                                                            |                  |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------|
| Production-Ready Hardware Design & Educational Measurement Kit<br>High-Frequency 1kHz Kelvin 4-Wire EIS Sensing Architecture & Ultra-Dense Via Arrays<br>Kit 2: PCB Track Resistance   4S1P (I) Differential ACIR   4S1P (II) Varying Interconnect IR<br>Comprehensive 18650 ACIR Characterization Platform<br><b>Decibels Lab Pvt Ltd</b> |                  |           |
| Sheet: /<br>File: ACIR-Learning-Kit-2-PCB.kicad_sch                                                                                                                                                                                                                                                                                        |                  |           |
| <b>Title: ACIR LEARNING KIT 2 – CHARACTERIZATION &amp; EXPERIMENTATION BOARD</b>                                                                                                                                                                                                                                                           |                  |           |
| Size: A3                                                                                                                                                                                                                                                                                                                                   | Date: 2026-09-16 | Rev: v2.0 |
| KiCad E.D.A. 10.0.5                                                                                                                                                                                                                                                                                                                        |                  | Id: 1/1   |